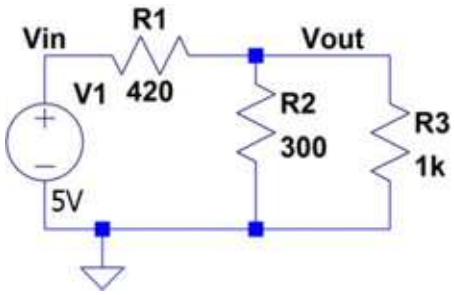
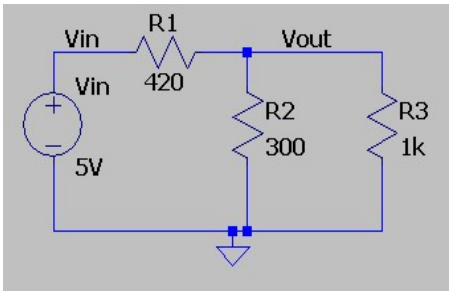
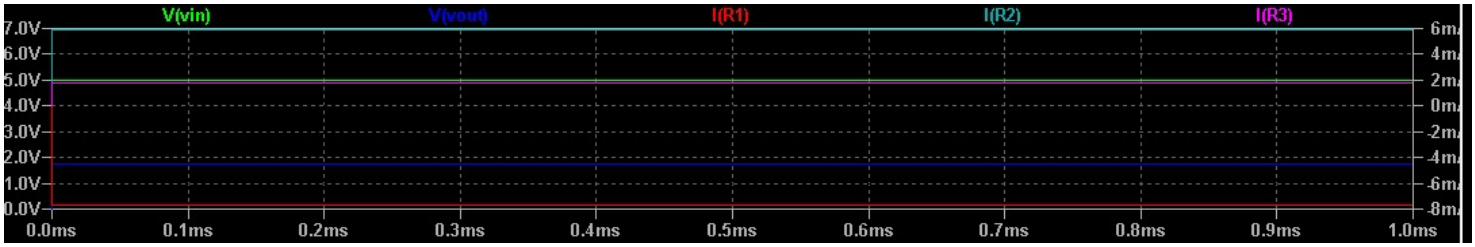
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2. Simulate the following circuit for 1ms, and compare the output voltage and current through each resistor with the results in Lab 1.



a) LTspice simulation:





B) Demonstration

$R_1 := 420\ \text{ohm}$ $R_2 := 300\ \text{ohm}$ $R_3 := 1\ \text{kohm}$ $R_{23} := \frac{1}{\left(\frac{1}{R_2} + \frac{1}{R_3}\right)} = 230.8\ \Omega$

$V_{in} := 5\ \text{volt}$

$V_{out} := V_{in} \cdot \frac{R_{23}}{R_1 + R_{23}} = 1.773\ \text{V}$

$I_{R1} := \frac{V_{out}}{R_1} = 0.004222\ \text{A}$ $I_{R2} := \frac{V_{out}}{R_2} = 0.005910\ \text{A}$ $I_{R3} := \frac{V_{out}}{R_3} = 0.001773\ \text{A}$

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